

CardioScan AI

End-to-End ECG Processing & Inference Pipeline

Training pipeline

MIT-BIH Arrhythmia Database 48 records • 360 Hz • two-channel ECG
WFDB record loading Read .dat + .hea; select MLII
Bandpass filter 4th-order Butterworth • 0.5–45 Hz • filtfilt
Training beat centers Use expert .atr annotation sample locations
Beat extraction 200 ms before + 380 ms after R-peak
Fixed-length input Pad/truncate to 280 samples
Per-beat normalization Z-score: (beat – mean) / std
Dataset split Stratified beat-level split • ~70/15/15
1D residual-style CNN 32 → 64 → 128 → 256 filters • dilations 1/2/4
Training Adam • 1e-3 initial LR • class weights • early stopping
Evaluation 98.95% accuracy • 96.91% macro F1

Inference / deployment pipeline

User upload Matching WFDB .dat + .hea files
Flask /api/analyze Save upload to temporary directory
WFDB read Read signal and sampling frequency
Resampling if required Inference path can resample to 360 Hz
Bandpass filter 4th-order Butterworth • 0.5–45 Hz
R-peak detection Derivative → square → 0.15 s moving average → adaptive threshold → 0.3 s minimum gap
Beat extraction Same 200 ms + 380 ms window; fixed 280 samples
Z-score normalization Per-beat normalization consistent with training
Model inference Reshape (N, 280, 1) → softmax(5)
Confidence / guard Mean confidence; low-confidence flag; Q-dominance warning
JSON response ECG signal • R-peaks • events • HR • rhythm • overall status
React results view Waveform canvas • event table • summary metrics

Important corrections

- Model parameter count: approximately 530,373 trainable parameters — not ~180K.
- Evaluation is beat-level stratified, not patient-independent.
- The preprocessing AAMI map contains one E→S mapping; the earlier duplicate E key was removed.
- Clinical claims and unsupported processing-time claims are not part of this corrected pipeline document.

Research prototype only. Not clinically validated and not intended for diagnosis or treatment decisions.